For M/V XIANG RUI KOU only

Rev.	Alteration or remark	Date	Drawn	Appr.

DELTAMARIN	Drawn: MVa / 04.07.2011 Checked: PJA / 04.07.2011 Appvd.: PJA / 04.07.2011	Scale Metr.	COSCO 50000 DWT SSHLV
Name: FINAL LOADING MANUAL ADDENDUM 1		Size: A4	Revision: Page: 2+21 Drawing No: B3598.1174.102 Drawing No: SC-101-01JS

File name:

CONTENTS

1 General

This document includes intact stability calculations for vessel carrying buoyant cargo Super Gorilla 30000t.

It is assumed that the buoyant cargo belongs to the vessel's stability hull, and therefore the buoyancy of the cargo is taken into account in the stability calculations.

This document is an additional part for loading manual. The document does not include all relevant background information relating operation of the vessel in presented conditions. Fore that reason this addendum should always be used together with the loading manual.

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2 Loading conditions

2.1 Summary of loading conditions

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CASE	TEXT	DISP t	DWT t	BW t	CARGO t	T m	TR m	HEEL degree	KMT m
LC21	Buoyant Cargo (Super Gorilla), departure	69163.9	48292.5	13871.4	30000.0	9.680	0.000	0.0	22.183
LC22	Buoyant Cargo (Super Gorilla), arrival	68491.8	47620.4	16356.2	30000.0	9.600	0.000	0.0	22.257

CASE	KGF m	GM m	GMCORR m	BMMIN tm	BMMAX tm	RELBM %	XRELBM m	SHMIN t	SHMAX t	RELSH %	XRELSH m
LC21	20.67	1.509	0.146	-131502	17172	73.7	109	-5611	4558	82.6	61
LC22	20.79	1.472	0.159	-83160	41573	49.3	115	-3974	4028	61.3	151

CASE	name	
TEXT	descriptive name	
DISP	total displacement	t
DWT	mass of load	t
BW	mass of load	t
CARGO	mass of load	t
T	draught, moulded	m
TR	trim	m
HEEL	heeling angle	degree
KMT	transv. metac. height	m
KGF	cgz of load	m
GM	metacentric height	m
GMCORR	GM correction	m
BMMIN	minimum bending moment	tm
BMMAX	maximum bending moment	tm
RELBM	relative bending moment	%
XRELBM	x where max. relative bending mom.	m
SHMIN	minimum shear force	t
SHMAX	maximum shear force	t
RELSH	relative shear force	%
XRELSH	x where max. relative shear force	m

2.2 LC21 Buoyant cargo (Super Gorilla), departure

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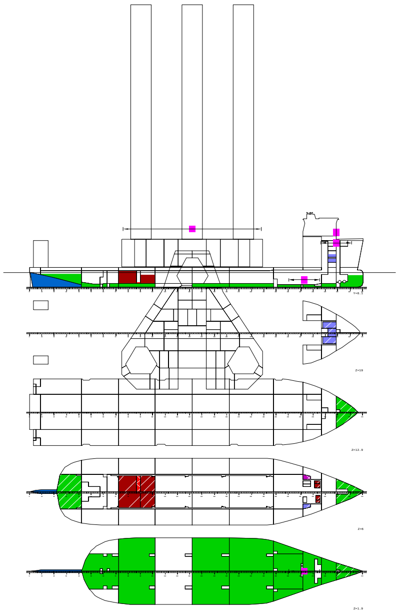


LOADING CONDITION LC21: Buoyant Cargo (Super Gorilla), departure

FLOATING POSITION

Mean draught (moulded)	9.68 m	KM above the moulded base	22.18 m
Draught at AP (moulded)	9.68 m	KG above the moulded base	20.53 m
Draught at FP (moulded)	9.68 m	GM0 (solid)	1.66 m
Trim	0.00 m	Free surface correction	-0.15 m
Heeling	0.0 deg		
		GM (fluid)	1.51 m
Mean draught (heel 0)	9.68 m		
Trim (heel 0)	0.00 m		

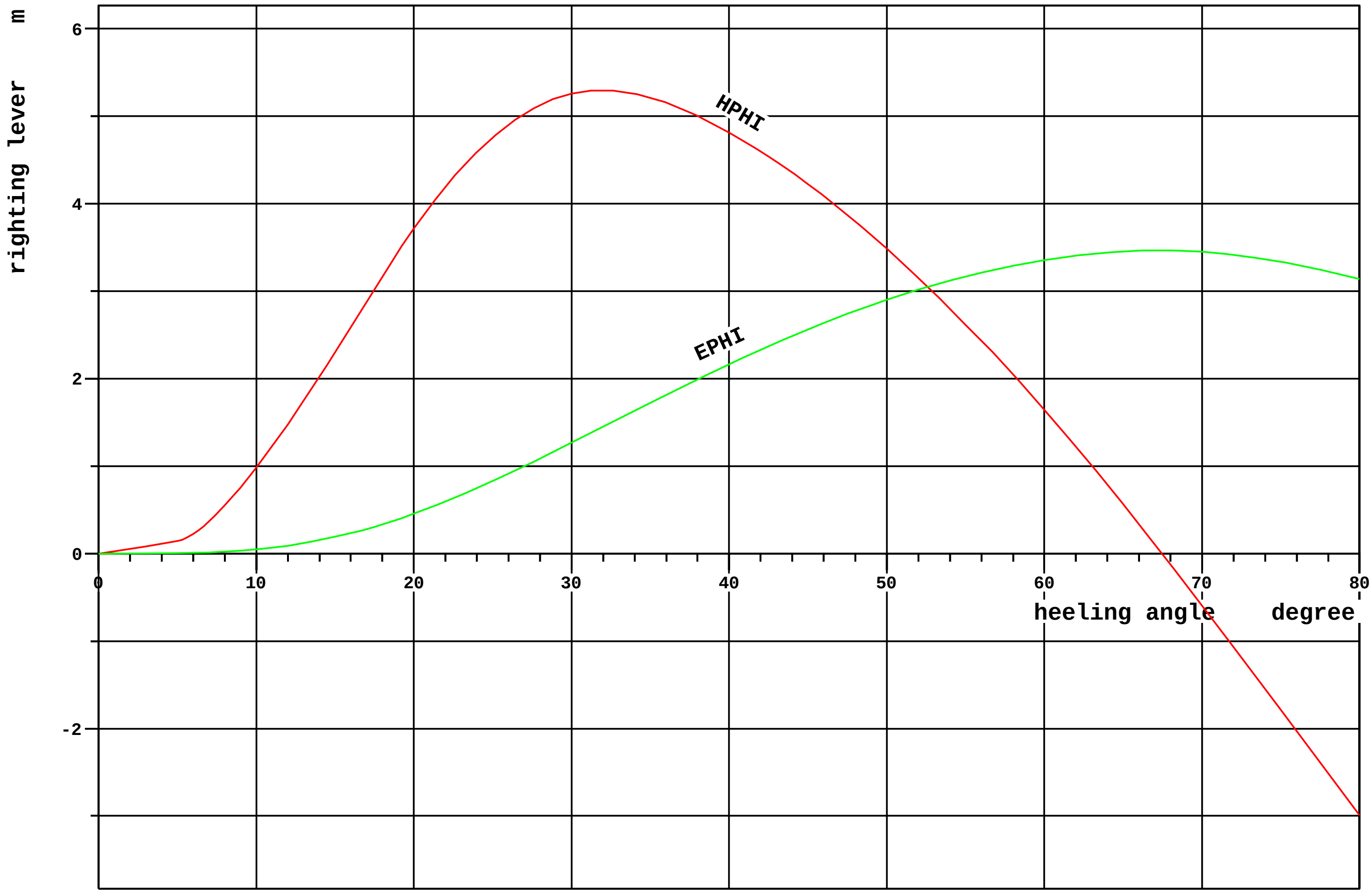
LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	13871.4	111.25	0.09	2.13	7798.8
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Solid cargo	30000.0	105.60	0.00	38.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	48292.5	105.14	0.00	24.88	10118.3
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	69163.9	107.65	-0.00	20.53	10118.3

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
_							
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HFO2	HFO Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HFO3	HFO Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HFO4	HFO Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	986.1	64.7	24.47	5.40	6.67	2635.3
WB02.06	Water Ballast 02.06	746.1	48.9	25.73	-5.41	6.10	2743.8
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	811.1	49.5	205.80	0.00	4.56	2419.7
Total of Ballast Water		13871.4		111.25	0.09	2.13	7798.8
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	30000.0		105.60	0.00	38.00	0.0

Location Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Stores (STO)	100.0		199.20	0.00	29.05	0.0
Crew (CREW)	6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3 (MIS)	87.0		178.40	0.00	5.00	0.0
Deadweight	48292.5		105.14	0.00	24.88	10118.3
Lightweight	20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)	69163.9		107.65	-0.00	20.53	10118.3

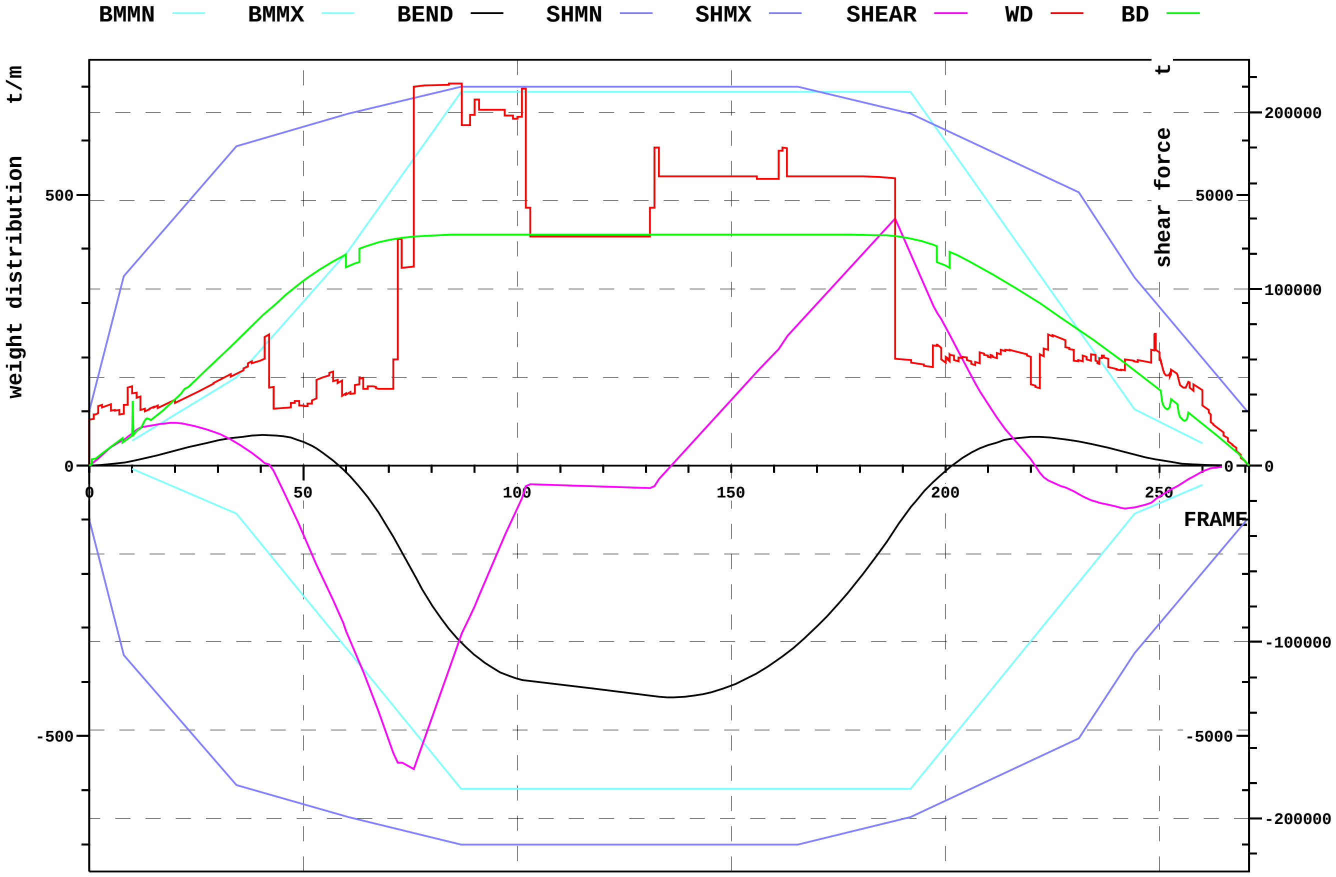


RCR	TEXT	REQ	ATTV	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	1.269	mrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.900	mrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	5.296	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	31.910	deg	OK
V.GM0.15	GM > 0.15 m	0.150	1.509	m	OK
IMOWEATHER	IMO weather criterion	1.000	7.698		OK
ND-RANGE	Range of Stability acc. ND	36.000	67.462	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	2.446		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	75.842	deg	OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	23.670
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	23.670
AIR_DUCT_1	UNPROTECTED	75.8	222.00	11.375	33.350	23.670

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

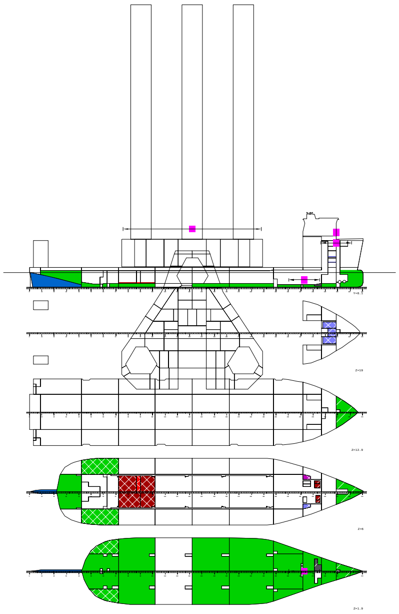
			x	frame
Shear force (min)	-5610.8 t	position:	60.6 m	76
Shear force (max)	4558.2 t		150.6 m	188
Max. rel. shear force	82.6 %		60.6 m	76
Sagging moment	-131502 tm		108.6 m	136
Hogging moment	17171.9 tm		33.6 m	42
Max. rel. bending moment	73.7 %		108.6 m	136

LOADING CONDITION LC21, Buoyant Cargo (Super Gorilla), departure

FR	X	BEND	BMMN	BMMX	RELBM
#	m	tm	tm	tm	%
21.00	16.80	9005	-13513	30261	2.9
42.00	33.60	17172	-50240	70924	11.3
57.00	45.60	2716	-94804	111952	5.7
72.00	57.60	-44685	-139384	160929	36.9
87.00	69.60	-100534	-183412	211763	58.1
102.00	81.60	-121756	-183412	211763	68.8
117.00	93.60	-126143	-183412	211763	71.0
132.00	105.60	-130957	-183412	211763	73.5
147.00	117.60	-127275	-183412	211763	71.6
162.00	129.60	-108129	-183412	211763	61.9
180.00	144.00	-63826	-183412	211763	39.5
198.00	158.40	-7398	-164872	190357	11.3
210.00	168.00	11515	-129204	149176	1.1
222.00	177.60	16204	-93541	108000	8.9
237.00	189.60	10609	-48977	56548	12.9
249.00	199.20	3448	-22505	25984	7.0

FR	X	SHEAR	SHMN	SHMX	RELSH
#	m	t	t	t	%
21.00	16.80	777	-4679	4679	16.6
42.00	33.60	14	-6075	6075	0.2
57.00	45.60	-2488	-6418	6418	38.8
72.00	57.60	-5487	-6718	6718	81.7
87.00	69.60	-3108	-7000	7000	44.4
102.00	81.60	-387	-7000	7000	5.5
117.00	93.60	-383	-7000	7000	5.5
132.00	105.60	-380	-7000	7000	5.4
147.00	117.60	954	-7000	7000	13.6
162.00	129.60	2262	-7000	7000	32.3
180.00	144.00	3850	-6724	6724	57.3
198.00	158.40	2823	-6269	6269	45.0
210.00	168.00	1129	-5824	5824	19.4
222.00	177.60	-138	-5380	5380	2.6
237.00	189.60	-709	-4339	4339	16.3
249.00	199.20	-634	-3024	3024	21.0

2.3 LC22 Buoyant cargo (Super Gorilla), arrival



LOADING CONDITION LC22: Buoyant Cargo (Super Gorilla), arrival

FLOATING POSITION

Mean draught (moulded)	9.60 m	KM above the moulded base	22.26 m
Draught at AP (moulded)	9.60 m	KG above the moulded base	20.63 m
Draught at FP (moulded)	9.60 m	GM0 (solid)	1.63 m
Trim	0.00 m	Free surface correction	-0.16 m
Heeling	0.0 deg		
		GM (fluid)	1.47 m
Mean draught (heel 0)	9.60 m		
Trim (heel 0)	0.00 m		

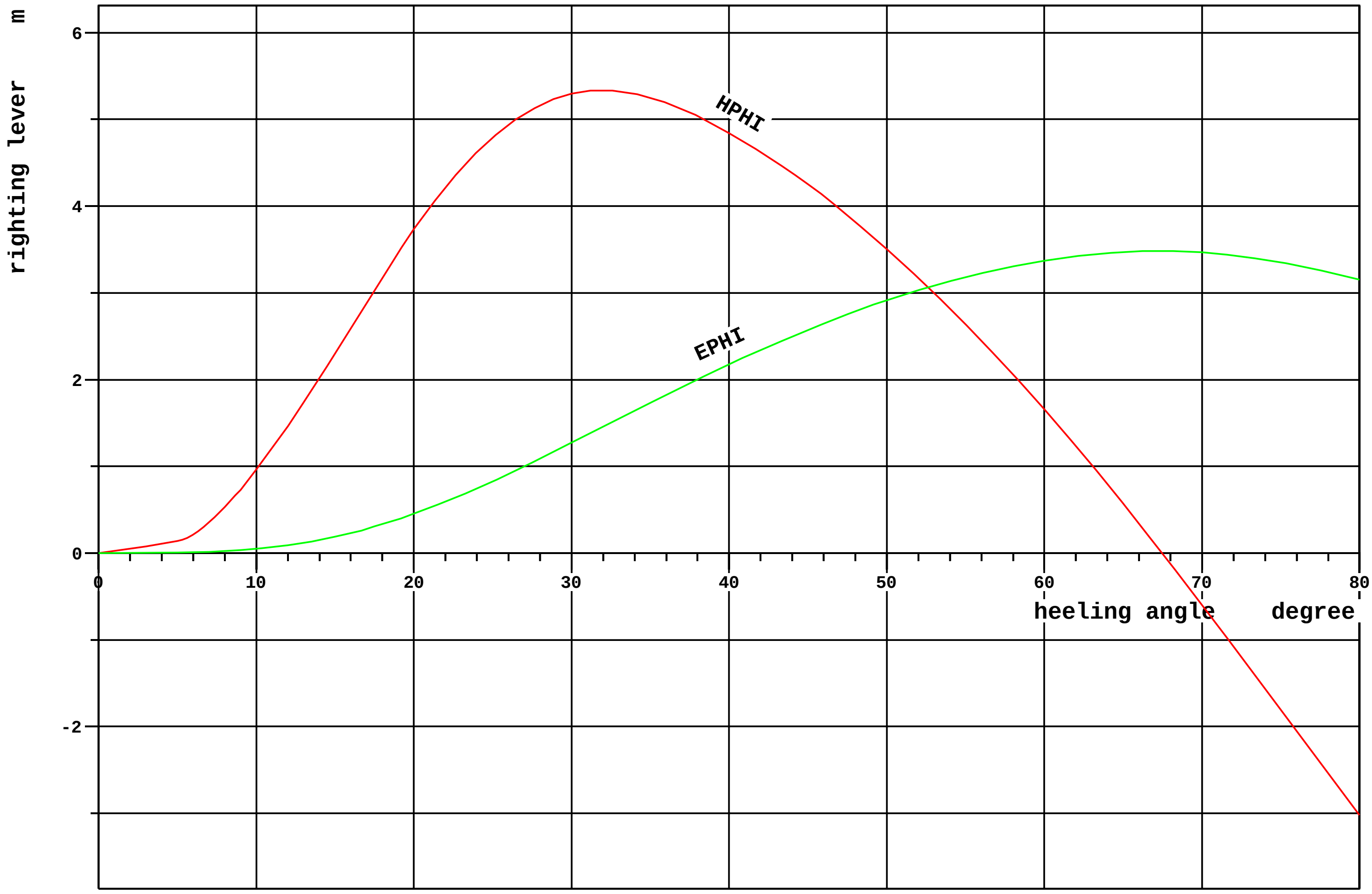
LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	16356.2	105.17	-0.01	3.00	5124.3
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Solid cargo	30000.0	105.60	0.00	38.00	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	47620.4	105.21	0.00	25.09	10866.1
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	68491.8	107.72	-0.00	20.63	10866.1

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
_							
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HFO2	HFO Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HFO3	HFO Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HFO4	HFO Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	1524.8	100.0	23.01	5.35	7.78	0.0
WB02.06	Water Ballast 02.06	1526.6	100.0	23.02	-5.35	7.78	0.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	282.1	13.3	50.39	15.06	1.85	1170.3
WB03.08	Water Ballast 03.08	292.8	13.8	50.30	-15.07	1.89	1197.3
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	1401.8	85.5	205.60	-0.01	6.71	2756.8
Total of Ballast Water		16356.2		105.17	-0.01	3.00	5124.3
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	30000.0		105.60	0.00	38.00	0.0
Stores							
(STO)		10.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		47620.4		105.21	0.00	25.09	10866.1
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		68491.8		107.72	-0.00	20.63	10866.1



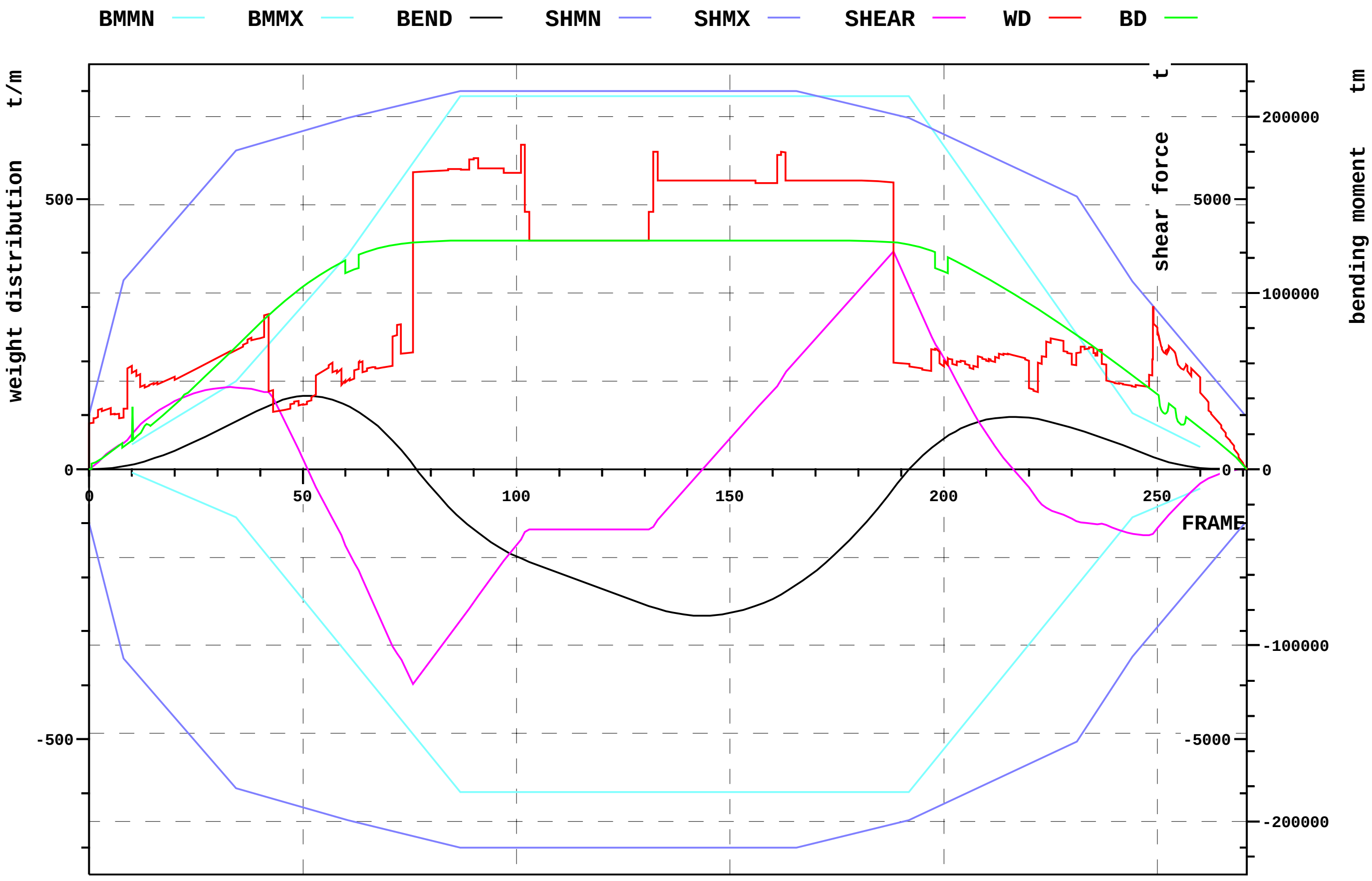
RCR	TEXT	REQ	ATTV	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	1.271	mrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.906	mrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	5.334	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	31.889	deg	OK
V.GM0.15	GM > 0.15 m	0.150	1.472	m	OK
IMOWEATHER	IMO weather criterion	1.000	7.697		OK
ND-RANGE	Range of Stability acc. ND	36.000	67.451	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	2.432		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	76.053	deg	OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	23.750
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	23.750
AIR_DUCT_1	UNPROTECTED	76.1	222.00	11.375	33.350	23.750

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m

Loading Condition
LC22



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

			x	frame
Shear force (min)	-3974.2 t	position:	60.6 m	76
Shear force (max)	4028.0 t		150.6 m	188
Max. rel. shear force	61.3 %		150.6 m	188
Sagging moment	-83160.0 tm		114.8 m	144
Hogging moment	41573.3 tm		40.8 m	51
Max. rel. bending moment	49.3 %		114.8 m	144

LOADING CONDITION LC22, Buoyant Cargo (Super Gorilla), arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	11581	-13513	30261	14.7
42.00	33.60	36090	-50240	70924	42.5
57.00	45.60	39326	-94804	111952	29.7
72.00	57.60	13534	-139384	160929	1.8
87.00	69.60	-27952	-183412	211763	21.3
102.00	81.60	-51694	-183412	211763	33.3
117.00	93.60	-65097	-183412	211763	40.1
132.00	105.60	-78418	-183412	211763	46.9
147.00	117.60	-82731	-183412	211763	49.0
162.00	129.60	-71070	-183412	211763	43.1
180.00	144.00	-35077	-183412	211763	24.9
198.00	158.40	13777	-164872	190357	0.6
210.00	168.00	28046	-129204	149176	13.0
222.00	177.60	28400	-93541	108000	21.0
237.00	189.60	17931	-48977	56548	26.8
249.00	199.20	6915	-22505	25984	21.3

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1295	-4679	4679	27.7
42.00	33.60	1432	-6075	6075	23.6
57.00	45.60	-913	-6418	6418	14.2
72.00	57.60	-3409	-6718	6718	50.7
87.00	69.60	-2796	-7000	7000	39.9
102.00	81.60	-1162	-7000	7000	16.6
117.00	93.60	-1115	-7000	7000	15.9
132.00	105.60	-1070	-7000	7000	15.3
147.00	117.60	307	-7000	7000	4.4
162.00	129.60	1658	-7000	7000	23.7
180.00	144.00	3297	-6724	6724	49.0
198.00	158.40	2320	-6269	6269	37.0
210.00	168.00	660	-5824	5824	11.3
222.00	177.60	-577	-5380	5380	10.7
237.00	189.60	-1014	-4339	4339	23.4
249.00	199.20	-1196	-3024	3024	39.5

3 KG_F-T limit curves

KG-T Summary, Trim 0
Vessel with buoyant cargo Super Gorilla

Proj	P6047/0
Date	2011-06-30
Time	17.02
Sign	MVA

Wind profile: SG90-PROFILE
Stability Hull: STABHULL+SG90